

Name: Monisha Raghini Chelle

Roll No: 2520030537

Branch: B.Tech CSE

Subject: Linux System Programming Lab

OSSP_Practical9&10

OSSP (Operating Systems And Systems Programming) - 25CS2104E

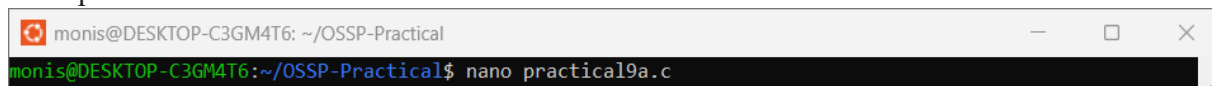
9.Implement a file copy utility using low-level Linux file I/O system calls (open, read, write, lseek, close). Compare its performance with standard library functions (fopen, fread, fwrite).

Develop a program that redirects standard input and output using file descriptors and dup2(). Demonstrate how shells implement I/O redirection.

Step 1 — Create practical9a.c

`cd ~/OSSP-Practical`

`nano practical9a.c`



```
monis@DESKTOP-C3GM4T6: ~/OSSP-Practical
GNU nano 8.7.1 practical9a.c *
#define _POSIX_C_SOURCE 200809L

#include <stdio.h>
#include <stdlib.h>
#include <fcntl.h>
#include <unistd.h>
#include <time.h>
#include <errno.h>

#define BUFFER_SIZE 8192

double elapsed(struct timespec start, struct timespec end)
{
    return (end.tv_sec - start.tv_sec) +
        (end.tv_nsec - start.tv_nsec) / 1e9;
}

int copy_syscall(const char *source, const char *destination)
{
    char buffer[BUFFER_SIZE];

    int in = open(source, O_RDONLY);
    if (in == -1) {
        perror("open source");
        return -1;
    }

    int out = open(destination, O_WRONLY | O_CREAT | O_TRUNC, 0644);
    if (out == -1) {
        perror("open destination");
        close(in);
        return -1;
    }

    /* Demonstrate lseek by positioning at the file beginning. */
    if (lseek(in, 0, SEEK_SET) == -1) {
        perror("lseek");
        close(in);
        close(out);
        return -1;
    }

    ssize_t n;

    while ((n = read(in, buffer, sizeof(buffer))) > 0) {

```

^G Help	^O Write Out	^F Where Is	^K Cut	^T Execute	^C Location	M-U Undo
^X Exit	^R Read File	^N Replace	^U Paste	^J Justify	^_ Go To Line	M-E Redo

```
monis@DESKTOP-C3GM4T6: ~/OSSP-Practical
GNU nano 8.7.1 practical9a.c *
while ((n = read(in, buffer, sizeof(buffer))) > 0) {
    ssize_t written = 0;

    while (written < n) {
        ssize_t result =
            write(out, buffer + written, (size_t)(n - written));

        if (result == -1) {
            if (errno == EINTR)
                continue;

            perror("write");
            close(in);
            close(out);
            return -1;
        }

        written += result;
    }
}

if (n == -1) {
    perror("read");
    close(in);
    close(out);
    return -1;
}

close(in);
close(out);

return 0;
}

int copy_stdio(const char *source, const char *destination)
{
    char buffer[BUFFER_SIZE];

    FILE *in = fopen(source, "rb");
    if (in == NULL) {
        perror("fopen source");
        return -1;
    }

    FILE *out = fopen(destination, "wb");
```

^G Help ^O Write Out ^F Where Is ^K Cut ^T Execute ^C Location M-U Undo
^X Exit ^R Read File ^V Replace ^U Paste ^J Justify ^_ Go To Line M-F Redo

```
Select monis@DESKTOP-C3GM4T6: ~/OSSP-Practical
GNU nano 8.7.1 practical9a.c *

FILE *out = fopen(destination, "wb");
if (out == NULL) {
    perror("fopen destination");
    fclose(in);
    return -1;
}

size_t n;

while ((n = fread(buffer, 1, sizeof(buffer), in)) > 0) {
    if (fwrite(buffer, 1, n, out) != n) {
        perror("fwrite");
        fclose(in);
        fclose(out);
        return -1;
    }
}

if (ferror(in)) {
    perror("fread");
    fclose(in);
    fclose(out);
    return -1;
}

fclose(in);
fclose(out);

return 0;
}

int main(int argc, char *argv[])
{
    if (argc != 2) {
        fprintf(stderr, "Usage: %s source_file\n", argv[0]);
        return 1;
    }

    struct timespec start, end;

    clock_gettime(CLOCK_MONOTONIC, &start);

    if (copy_syscall(argv[1], "copy_syscall.txt") != 0)
```

```
monis@DESKTOP-C3GM4T6: ~/OSSP-Practical
GNU nano 8.7.1 practical9a.c *
    fclose(in);
    fclose(out);
    return -1;
}

fclose(in);
fclose(out);

return 0;
}

int main(int argc, char *argv[])
{
    if (argc != 2) {
        fprintf(stderr, "Usage: %s source_file\n", argv[0]);
        return 1;
    }

    struct timespec start, end;

    clock_gettime(CLOCK_MONOTONIC, &start);

    if (copy_syscall(argv[1], "copy_syscall.txt") != 0)
        return 1;

    clock_gettime(CLOCK_MONOTONIC, &end);

    printf("System call copy time: %.6f seconds\n",
        elapsed(start, end));

    clock_gettime(CLOCK_MONOTONIC, &start);

    if (copy_stdio(argv[1], "copy_stdio.txt") != 0)
        return 1;

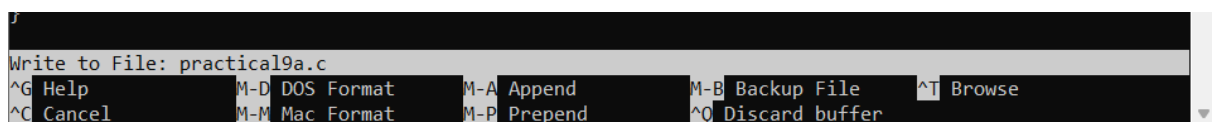
    clock_gettime(CLOCK_MONOTONIC, &end);

    printf("Standard library copy time: %.6f seconds\n",
        elapsed(start, end));

    printf("Both file copies completed successfully.\n");

    return 0;
}
```

Save with Ctrl + O, Enter, then Ctrl + X.



Step 2 — Create a test file

For a quick test:

```
echo "OSSP Practical 9 File Copy Test" > practical9_input.txt
```

```
monis@DESKTOP-C3GM4T6: ~/OSSP-Practical
monis@DESKTOP-C3GM4T6:~/OSSP-Practical$ nano practical9a.c
monis@DESKTOP-C3GM4T6:~/OSSP-Practical$ echo "OSSP Practical 9 File Copy Test" > practical9_input.txt
monis@DESKTOP-C3GM4T6:~/OSSP-Practical$
```

Step 3 — Compile and run

```
gcc -Wall -Wextra practical9a.c -o practical9a
./practical9a practical9_input.txt
```

```
Select monis@DESKTOP-C3GM4T6: ~/OSSP-Practical
monis@DESKTOP-C3GM4T6:~/OSSP-Practical$ nano practical9a.c
monis@DESKTOP-C3GM4T6:~/OSSP-Practical$ echo "OSSP Practical 9 File Copy Test" > practical9_input.txt
monis@DESKTOP-C3GM4T6:~/OSSP-Practical$ gcc -Wall -Wextra practical9a.c -o practical9a
monis@DESKTOP-C3GM4T6:~/OSSP-Practical$ ./practical9a practical9_input.txt
System call copy time: 0.000209 seconds
Standard library copy time: 0.001730 seconds
Both file copies completed successfully.
monis@DESKTOP-C3GM4T6:~/OSSP-Practical$
```

Step 4 — Verify both copies

```
cat copy_syscall.txt
cat copy_stdio.txt
```

```
cmp practical9_input.txt copy_syscall.txt
cmp practical9_input.txt copy_stdio.txt
```

```
monis@DESKTOP-C3GM4T6: ~/OSSP-Practical
monis@DESKTOP-C3GM4T6:~/OSSP-Practical$ nano practical9a.c
monis@DESKTOP-C3GM4T6:~/OSSP-Practical$ echo "OSSP Practical 9 File Copy Test" > practical9_input.txt
monis@DESKTOP-C3GM4T6:~/OSSP-Practical$ gcc -Wall -Wextra practical9a.c -o practical9a
monis@DESKTOP-C3GM4T6:~/OSSP-Practical$ ./practical9a practical9_input.txt
System call copy time: 0.000209 seconds
Standard library copy time: 0.001730 seconds
Both file copies completed successfully.
monis@DESKTOP-C3GM4T6:~/OSSP-Practical$ cat copy_syscall.txt
OSSP Practical 9 File Copy Test
monis@DESKTOP-C3GM4T6:~/OSSP-Practical$ cat copy_stdio.txt
OSSP Practical 9 File Copy Test
monis@DESKTOP-C3GM4T6:~/OSSP-Practical$ cmp practical9_input.txt copy_syscall.txt
monis@DESKTOP-C3GM4T6:~/OSSP-Practical$ cmp practical9_input.txt copy_stdio.txt
monis@DESKTOP-C3GM4T6:~/OSSP-Practical$
```

To perform a more meaningful performance comparison, create a larger file:

```
dd if=/dev/zero of=practical9_large.bin bs=1M count=20 status=none
./practical9a practical9_large.bin
```

```
monis@DESKTOP-C3GM4T6: ~/OSSP-Practical
monis@DESKTOP-C3GM4T6:~/OSSP-Practical$ nano practical9a.c
monis@DESKTOP-C3GM4T6:~/OSSP-Practical$ echo "OSSP Practical 9 File Copy Test" > practical9_input.txt
monis@DESKTOP-C3GM4T6:~/OSSP-Practical$ gcc -Wall -Wextra practical9a.c -o practical9a
monis@DESKTOP-C3GM4T6:~/OSSP-Practical$ ./practical9a practical9_input.txt
System call copy time: 0.000209 seconds
Standard library copy time: 0.001730 seconds
Both file copies completed successfully.
monis@DESKTOP-C3GM4T6:~/OSSP-Practical$ cat copy_syscall.txt
OSSP Practical 9 File Copy Test
monis@DESKTOP-C3GM4T6:~/OSSP-Practical$ cat copy_stdio.txt
OSSP Practical 9 File Copy Test
monis@DESKTOP-C3GM4T6:~/OSSP-Practical$ cmp practical9_input.txt copy_syscall.txt
monis@DESKTOP-C3GM4T6:~/OSSP-Practical$ cmp practical9_input.txt copy_stdio.txt
monis@DESKTOP-C3GM4T6:~/OSSP-Practical$ dd if=/dev/zero of=practical9_large.bin bs=1M count=20 status=no
ne
./practical9a practical9_large.bin
System call copy time: 0.049439 seconds
Standard library copy time: 0.042121 seconds
Both file copies completed successfully.
monis@DESKTOP-C3GM4T6:~/OSSP-Practical$
```

Step 5 — Create practical9b.c
nano practical9b.c

```
monis@DESKTOP-C3GM4T6: ~/OSSP-Practical
monis@DESKTOP-C3GM4T6:~/OSSP-Practical$ nano practical9b.c
monis@DESKTOP-C3GM4T6:~/OSSP-Practical$
```

```
monis@DESKTOP-C3GM4T6: ~/OSSP-Practical
GNU nano 8.7.1 practical9b.c *
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <fcntl.h>
#include <sys/wait.h>

int main(void)

    pid_t pid = fork();

    if (pid < 0) {
        perror("fork");
        return 1;
    }

    if (pid == 0) {
        int input_fd = open("practical9_input.txt", O_RDONLY);

        if (input_fd == -1) {
            perror("open input");
            _exit(1);
        }

        int output_fd = open("practical9_output.txt",
                            O_WRONLY | O_CREAT | O_TRUNC,
                            0644);

        if (output_fd == -1) {
            perror("open output");
            close(input_fd);
            _exit(1);
        }

        if (dup2(input_fd, STDIN_FILENO) == -1) {
            perror("dup2 stdin");
            _exit(1);
        }

        if (dup2(output_fd, STDOUT_FILENO) == -1) {
            perror("dup2 stdout");
            _exit(1);
        }

        close(input_fd);
        close(output_fd);
    }

G Help      ^O Write Out ^F Where Is  ^K Cut      ^T Execute  ^C Location  M-U Undo
X Exit      ^R Read File ^\ Replace   ^U Paste    ^J Justify  ^_ Go To Line M-E Redo
```

```
monis@DESKTOP-C3GM4T6: ~/OSSP-Practical
GNU nano 8.7.1 practical9b.c *
0644);

    if (output_fd == -1) {
        perror("open output");
        close(input_fd);
        _exit(1);
    }

    if (dup2(input_fd, STDIN_FILENO) == -1) {
        perror("dup2 stdin");
        _exit(1);
    }

    if (dup2(output_fd, STDOUT_FILENO) == -1) {
        perror("dup2 stdout");
        _exit(1);
    }

    close(input_fd);
    close(output_fd);

    /* Equivalent to: cat < input.txt > output.txt */
    execlp("cat", "cat", (char *)NULL);

    perror("execlp");
    _exit(127);
}

int status;

if (waitpid(pid, &status, 0) == -1) {
    perror("waitpid");
    return 1;
}

if (WIFEXITED(status) && WEXITSTATUS(status) == 0) {
    printf("Input and output redirection successful.\n");
    printf("Output saved to practical9_output.txt\n");
    return 0;
}

printf("Redirection failed.\n");
return 1;
}

Write to File: practical9b.c
^G Help      ^O Write Out ^F Where Is  ^K Cut       ^T Execute   ^C Location  ^U Undo
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^/ Go To Line ^-E Redo

^G Help      ^M-D DOS Format ^M-A Append  ^M-B Backup File ^T Browse
^C Cancel     ^M-M Mac Format ^M-P Prepend ^O Discard buffer
```

Step 6 — Compile and run

```
gcc -Wall -Wextra practical9b.c -o practical9b
```

```
./practical9b
```

```
cat practical9_output.txt
```

```
monis@DESKTOP-C3GM4T6: ~/OSSP-Practical$ nano practical9b.c
monis@DESKTOP-C3GM4T6:~/OSSP-Practical$ gcc -Wall -Wextra practical9b.c -o practical9b
monis@DESKTOP-C3GM4T6:~/OSSP-Practical$ ./practical9b
Input and output redirection successful.
Output saved to practical9_output.txt
monis@DESKTOP-C3GM4T6:~/OSSP-Practical$ cat practical9_output.txt
OSSP Practical 9 File Copy Test
monis@DESKTOP-C3GM4T6:~/OSSP-Practical$
```

Step 7 — Compare with shell redirection

Run:

```
cat < practical9_input.txt > shell_output.txt
cmp practical9_output.txt shell_output.txt
echo $?
```

```
monis@DESKTOP-C3GM4T6: ~/OSSP-Practical$ nano practical9b.c
monis@DESKTOP-C3GM4T6:~/OSSP-Practical$ gcc -Wall -Wextra practical9b.c -o practical9b
monis@DESKTOP-C3GM4T6:~/OSSP-Practical$ ./practical9b
Input and output redirection successful.
Output saved to practical9_output.txt
monis@DESKTOP-C3GM4T6:~/OSSP-Practical$ cat practical9_output.txt
OSSP Practical 9 File Copy Test
monis@DESKTOP-C3GM4T6:~/OSSP-Practical$ cat < practical9_input.txt > shell_output.txt
monis@DESKTOP-C3GM4T6:~/OSSP-Practical$ cmp practical9_output.txt shell_output.txt
monis@DESKTOP-C3GM4T6:~/OSSP-Practical$ echo $?
0
monis@DESKTOP-C3GM4T6:~/OSSP-Practical$
```

10. Investigate inode structures using commands such as `ls -li`, `stat`, and `find`. Create hard links and symbolic links and explain their impact on inode allocation.

Design a program using `mmap()` to perform file reading and writing. Compare memory-mapped I/O with traditional read/write operations in terms of efficiency and implementation complexity.

Step 1 — Create a test file and its links

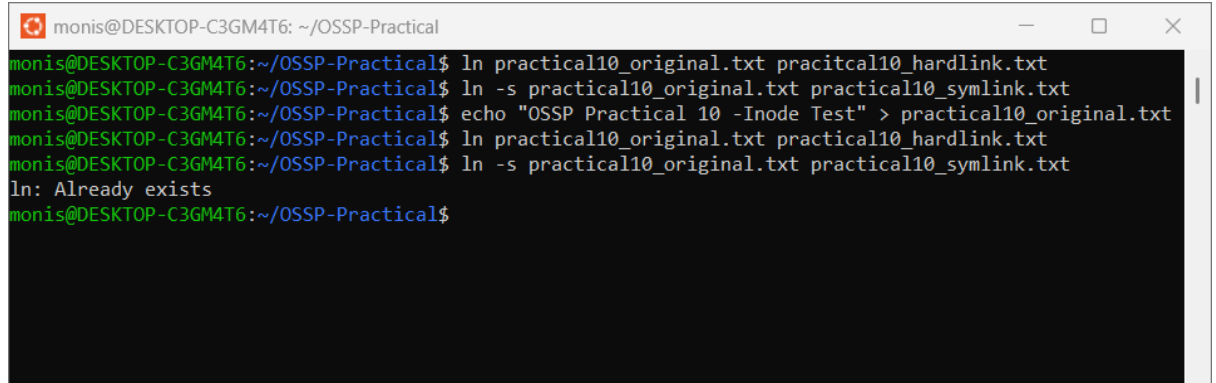
Run these commands in Ubuntu:

```
cd ~/OSSP-Practical
```

```
echo "OSSP Practical 10 - Inode Test" > practical10_original.txt
```

```
ln practical10_original.txt practical10_hardlink.txt
```

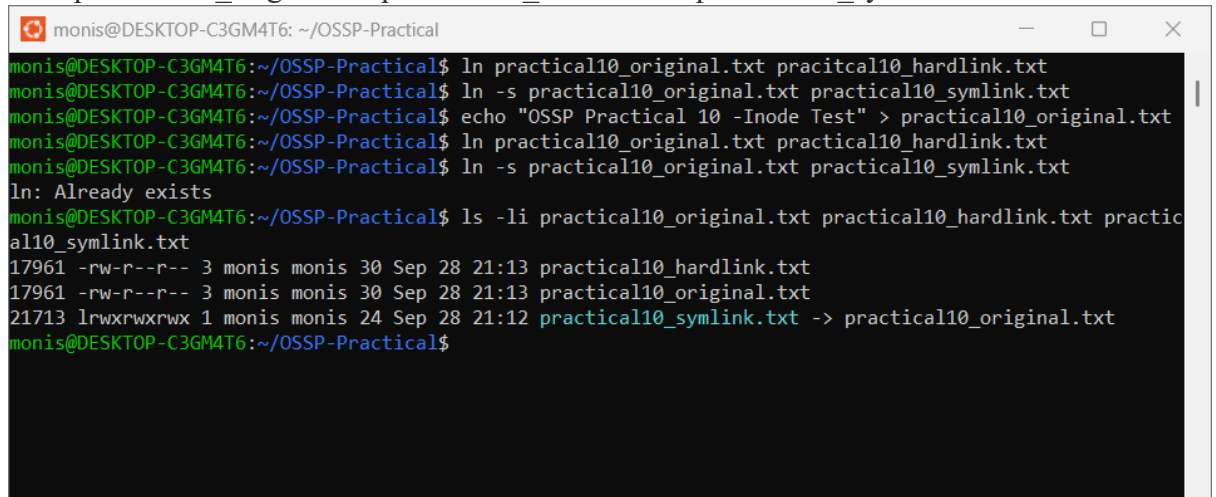
```
ln -s practical10_original.txt practical10_symlink.txt
```

A terminal window titled 'monis@DESKTOP-C3GM4T6: ~/OSSP-Practical' showing a series of commands to create links. The commands are: 'ln practical10_original.txt practical10_hardlink.txt', 'ln -s practical10_original.txt practical10_symlink.txt', 'echo "OSSP Practical 10 -Inode Test" > practical10_original.txt', 'ln practical10_original.txt practical10_hardlink.txt', and 'ln -s practical10_original.txt practical10_symlink.txt'. The last command results in the message 'ln: Already exists'.

```
monis@DESKTOP-C3GM4T6: ~/OSSP-Practical
monis@DESKTOP-C3GM4T6:~/OSSP-Practical$ ln practical10_original.txt practical10_hardlink.txt
monis@DESKTOP-C3GM4T6:~/OSSP-Practical$ ln -s practical10_original.txt practical10_symlink.txt
monis@DESKTOP-C3GM4T6:~/OSSP-Practical$ echo "OSSP Practical 10 -Inode Test" > practical10_original.txt
monis@DESKTOP-C3GM4T6:~/OSSP-Practical$ ln practical10_original.txt practical10_hardlink.txt
monis@DESKTOP-C3GM4T6:~/OSSP-Practical$ ln -s practical10_original.txt practical10_symlink.txt
ln: Already exists
monis@DESKTOP-C3GM4T6:~/OSSP-Practical$
```

Step 2 — Display inode numbers

```
ls -li practical10_original.txt practical10_hardlink.txt practical10_symlink.txt
```

A terminal window titled 'monis@DESKTOP-C3GM4T6: ~/OSSP-Practical' showing the same commands as the previous screenshot, followed by 'ls -li practical10_original.txt practical10_hardlink.txt practical10_symlink.txt'. The output shows the inode numbers for each file: 17961 for practical10_hardlink.txt, 17961 for practical10_original.txt, and 21713 for practical10_symlink.txt. The symlink entry also shows the target file: 'practical10_symlink.txt -> practical10_original.txt'.

```
monis@DESKTOP-C3GM4T6:~/OSSP-Practical$ ln practical10_original.txt practical10_hardlink.txt
monis@DESKTOP-C3GM4T6:~/OSSP-Practical$ ln -s practical10_original.txt practical10_symlink.txt
monis@DESKTOP-C3GM4T6:~/OSSP-Practical$ echo "OSSP Practical 10 -Inode Test" > practical10_original.txt
monis@DESKTOP-C3GM4T6:~/OSSP-Practical$ ln practical10_original.txt practical10_hardlink.txt
monis@DESKTOP-C3GM4T6:~/OSSP-Practical$ ln -s practical10_original.txt practical10_symlink.txt
ln: Already exists
monis@DESKTOP-C3GM4T6:~/OSSP-Practical$ ls -li practical10_original.txt practical10_hardlink.txt practical10_symlink.txt
17961 -rw-r--r-- 3 monis monis 30 Sep 28 21:13 practical10_hardlink.txt
17961 -rw-r--r-- 3 monis monis 30 Sep 28 21:13 practical10_original.txt
21713 lrwxrwxrwx 1 monis monis 24 Sep 28 21:12 practical10_symlink.txt -> practical10_original.txt
monis@DESKTOP-C3GM4T6:~/OSSP-Practical$
```

Step 3 — Examine inode information

```
stat practical10_original.txt
```

```
stat practical10_hardlink.txt
```

```
stat practical10_symlink.txt
```

To find files sharing the original file's inode:

```
find . -maxdepth 1 -samefile practical10_original.txt
```

```
monis@DESKTOP-C3GM4T6: ~/OSSP-Practical
monis@DESKTOP-C3GM4T6:~/OSSP-Practical$ stat practical10_original.txt
stat practical10_hardlink.txt
stat practical10_symlink.txt
  File: practical10_original.txt
  size: 30          Blocks: 8          IO Block: 4096   regular file
Device: 8,48   Inode: 17961          Links: 3
Access: (0644/-rw-r--r--)  Uid: ( 1000/   monis)   Gid: ( 1000/   monis)
Access: 2026-09-28 21:11:49.864204603 +0530
Modify: 2026-09-28 21:13:15.020426605 +0530
Change: 2026-09-28 21:13:36.440494535 +0530
 Birth: 2026-09-28 21:11:49.864204603 +0530
  File: practical10_hardlink.txt
  size: 30          Blocks: 8          IO Block: 4096   regular file
Device: 8,48   Inode: 17961          Links: 3
Access: (0644/-rw-r--r--)  Uid: ( 1000/   monis)   Gid: ( 1000/   monis)
Access: 2026-09-28 21:11:49.864204603 +0530
Modify: 2026-09-28 21:13:15.020426605 +0530
Change: 2026-09-28 21:13:36.440494535 +0530
 Birth: 2026-09-28 21:11:49.864204603 +0530
  File: 'practical10_symlink.txt' -> 'practical10_original.txt'
  size: 24          Blocks: 0          IO Block: 4096   symbolic link
Device: 8,48   Inode: 21713          Links: 1
Access: (0777/lrwxrwxrwx)  Uid: ( 1000/   monis)   Gid: ( 1000/   monis)
Access: 2026-09-28 21:13:44.736517556 +0530
Modify: 2026-09-28 21:12:59.296376048 +0530
Change: 2026-09-28 21:12:59.296376048 +0530
 Birth: 2026-09-28 21:12:59.296376048 +0530
monis@DESKTOP-C3GM4T6:~/OSSP-Practical$ find . -maxdepth 1 -samefile practical10_original.txt
./practical10_hardlink.txt
./practical10_original.txt
./practical10_hardlink.txt
monis@DESKTOP-C3GM4T6:~/OSSP-Practical$
```

To inspect the symbolic link:
readlink practical10_symlink.txt

```
monis@DESKTOP-C3GM4T6: ~/OSSP-Practical
monis@DESKTOP-C3GM4T6:~/OSSP-Practical$ find . -maxdepth 1 -samefile practical10_original.txt
./practical10_hardlink.txt
./practical10_original.txt
./practical10_hardlink.txt
monis@DESKTOP-C3GM4T6:~/OSSP-Practical$ readlink practical10_symlink.txt
practical10_original.txt
monis@DESKTOP-C3GM4T6:~/OSSP-Practical$
```

Step 4 — Demonstrate link behavior

Remove only the test original file:

```
rm practical10_original.txt
```

```
cat practical10_hardlink.txt
```

```
cat practical10_symlink.txt
```

```
monis@DESKTOP-C3GM4T6: ~/OSSP-Practical
monis@DESKTOP-C3GM4T6:~/OSSP-Practical$ rm practical10_original.txt

cat practical10_hardlink.txt
cat practical10_symlink.txt
OSSP Practical 10 -Inode Test
cat: practical10_symlink.txt: No such file or directory
monis@DESKTOP-C3GM4T6:~/OSSP-Practical$
```

Restore the original file for your practical record:
`cp practical10_hardlink.txt practical10_original.txt`

```
monis@DESKTOP-C3GM4T6: ~/OSSP-Practical
./practical10_original.txt
./practical10_hardlink.txt
monis@DESKTOP-C3GM4T6:~/OSSP-Practical$ readlink practical10_symlink.txt
practical10_original.txt
monis@DESKTOP-C3GM4T6:~/OSSP-Practical$ rm practical10_original.txt

cat practical10_hardlink.txt
cat practical10_symlink.txt
OSSP Practical 10 -Inode Test
cat: practical10_symlink.txt: No such file or directory
monis@DESKTOP-C3GM4T6:~/OSSP-Practical$ cp practical10_hardlink.txt practical10_original.txt
monis@DESKTOP-C3GM4T6:~/OSSP-Practical$
```

Step 5 — Create practical10b.c
`nano practical10b.c`

```
monis@DESKTOP-C3GM4T6: ~/OSSP-Practical
monis@DESKTOP-C3GM4T6:~/OSSP-Practical$ nano practical10b.c
monis@DESKTOP-C3GM4T6:~/OSSP-Practical$
```

```
monis@DESKTOP-C3GM4T6: ~/OSSP-Practical
GNU nano 8.7.1 practical10b.c *
#define _POSIX_C_SOURCE 200809L

#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <fcntl.h>
#include <sys/mman.h>
#include <sys/stat.h>
#include <time.h>
#include <errno.h>

#define SIZE (10 * 1024 * 1024)
#define BUFFER_SIZE 8192

double elapsed(struct timespec start, struct timespec end)
{
    return (end.tv_sec - start.tv_sec) +
        (end.tv_nsec - start.tv_nsec) / 1e9;
}

int main(void)
{
    struct timespec start, end;
    char *message = "OSSP Practical 10 - mmap Test\n";

    /* Create a 10 MB test file. */
    int fd = open("practical10_mmap.dat",
        O_RDWR | O_CREAT | O_TRUNC, 0644);

    if (fd == -1) {
        perror("open");
        return 1;
    }

    if (ftruncate(fd, SIZE) == -1) {
        perror("ftruncate");
        close(fd);
        return 1;
    }

    /* Memory-mapped I/O */
    clock_gettime(CLOCK_MONOTONIC, &start);

    char *mapped = mmap(NULL, SIZE,
```

^G Help	^O Write Out	^F Where Is	^K Cut	^T Execute	^C Location	M-U Undo
^X Exit	^R Read File	^I Replace	^U Paste	^J Justify	^_ Go To Line	M-E Redo

```
monis@DESKTOP-C3GM4T6: ~/OSSP-Practical
GNU nano 8.7.1 practical10b.c *
char *mapped = mmap(NULL, SIZE,
                    PROT_READ | PROT_WRITE,
                    MAP_SHARED, fd, 0);

if (mapped == MAP_FAILED) {
    perror("mmap");
    close(fd);
    return 1;
}

memcpy(mapped, message, strlen(message));

printf("mmap read: %s", mapped);

if (msync(mapped, SIZE, MS_SYNC) == -1)
    perror("msync");

if (munmap(mapped, SIZE) == -1)
    perror("munmap");

clock_gettime(CLOCK_MONOTONIC, &end);

printf("mmap time: %.6f seconds\n",
        elapsed(start, end));

close(fd);

/* Traditional read/write I/O */
fd = open("practical10_mmap.dat", O_RDONLY);

if (fd == -1) {
    perror("open input");
    return 1;
}

int out = open("practical10_copy.dat",
               O_WRONLY | O_CREAT | O_TRUNC, 0644);

if (out == -1) {
    perror("open output");
    close(fd);
    return 1;
}

char buffer[BUFFER_SIZE];

^G Help      ^O Write Out ^F Where Is  ^K Cut       ^T Execute   ^C Location  M-U Undo
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^_ Go To Line M-E Redo
```

```
monis@DESKTOP-C3GM4T6: ~/OSSP-Practical
GNU nano 8.7.1 practical10b.c *
char buffer[BUFFER_SIZE];
ssize_t n;

clock_gettime(CLOCK_MONOTONIC, &start);

while ((n = read(fd, buffer, sizeof(buffer))) > 0) {
    ssize_t written = 0;

    while (written < n) {
        ssize_t result =
            write(out, buffer + written,
                (size_t)(n - written));

        if (result == -1) {
            if (errno == EINTR)
                continue;

            perror("write");
            close(fd);
            close(out);
            return 1;
        }

        if (result == 0) {
            fprintf(stderr, "Write made no progress.\n");
            close(fd);
            close(out);
            return 1;
        }

        written += result;
    }
}

if (n == -1) {
    perror("read");
    close(fd);
    close(out);
    return 1;
}

close(fd);
close(out);

clock_gettime(CLOCK_MONOTONIC, &end);

^G Help      ^O Write Out  ^F Where Is  ^K Cut        ^T Execute   ^C Location  M-U Undo
^X Exit      ^R Read File  ^\ Replace   ^U Paste      ^J Justify   ^_ Go To Line M-E Redo
```



```
monis@DESKTOP-C3GM4T6: ~/OSSP-Practical
GNU nano 8.7.1 practical10b.c *
    ssize_t result =
        write(out, buffer + written,
              (size_t)(n - written));

    if (result == -1) {
        if (errno == EINTR)
            continue;

        perror("write");
        close(fd);
        close(out);
        return 1;
    }

    if (result == 0) {
        fprintf(stderr, "Write made no progress.\n");
        close(fd);
        close(out);
        return 1;
    }

    written += result;
}

if (n == -1) {
    perror("read");
    close(fd);
    close(out);
    return 1;
}

close(fd);
close(out);

clock_gettime(CLOCK_MONOTONIC, &end);

printf("read/write copy time: %.6f seconds\n",
       elapsed(start, end));

printf("Both methods executed successfully.\n");

return 0;

; Help      ^O Write Out  ^F Where Is  ^K Cut       ^T Execute   ^C Location  M-U Undo
C Exit      ^R Read File  ^\ Replace   ^U Paste     ^J Justify   ^_ Go To Line M-E Redo

Write to File: practical10b.c
^G Help      M-D DOS Format  M-A Append    M-B Backup File ^T Browse
^C Cancel     M-M Mac Format  M-P Prepend    ^Q Discard buffer
```

Step 6 — Compile and run

```
gcc -Wall -Wextra practical10b.c -o practical10b
./practical10b
```

```
monis@DESKTOP-C3GM4T6: ~/OSSP-Practical
monis@DESKTOP-C3GM4T6:~/OSSP-Practical$ nano practical10b.c
monis@DESKTOP-C3GM4T6:~/OSSP-Practical$ gcc -Wall -Wextra practical10b.c -o practical10b
monis@DESKTOP-C3GM4T6:~/OSSP-Practical$ ./practical10b
mmap read: OSSP Practical 10 - mmap Test
mmap time: 0.019730 seconds
read/write copy time: 0.027655 seconds
Both methods executed successfully.
monis@DESKTOP-C3GM4T6:~/OSSP-Practical$
```

Step 7 — Verify file contents

```
head -c 30 practical10_mmap.dat
echo
```

```
head -c 30 practical10_copy.dat
echo
```

```
cmp practical10_mmap.dat practical10_copy.dat
echo $?
```

```
monis@DESKTOP-C3GM4T6: ~/OSSP-Practical
monis@DESKTOP-C3GM4T6:~/OSSP-Practical$ head -c 30 practical10_mmap.dat
echo

head -c 30 practical10_copy.dat
echo

cmp practical10_mmap.dat practical10_copy.dat
echo $?
OSSP Practical 10 - mmap Test

OSSP Practical 10 - mmap Test

0
monis@DESKTOP-C3GM4T6:~/OSSP-Practical$
```

Memory-mapped I/O	Traditional read/write
Uses <code>mmap()</code> to map a file into virtual memory.	Uses <code>read()</code> and <code>write()</code> to transfer data through a buffer.
File contents can be accessed through memory addresses.	File contents are accessed through explicit system calls.

Useful for random access and repeated access to mapped data.	Straightforward for sequential copying and streaming.
Requires careful mapping, synchronization and unmapping.	Requires handling partial reads and writes.
Can reduce explicit copying in some workloads.	Performance depends on buffering, file size and workload.

Result: Inode numbers and link behavior were investigated. Memory-mapped file access was implemented and compared with traditional file I/O.